Z-B-421E
August 10, 1971
SUPERSEDING
(See 6.4)

FEDERAL SPECIFICATION

BERRIES, CANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for canned berries for use by all Federal agencies.

1.2 Classification. The canned berries shall be of the following types and kinds, as specified (see 6.1).

Type I. Blackberries and similar berries

Kinds (A.) Blackberries
(B.) Boysenberries
(C.) Dewberries
(D.) Loganberries
(E.) Youngberries

Type II. Blueberries

Kinds (A.) Native (or wild)
(B.) Cultivated

Type III. Raspberries

Kinds (A.) Red (other than reddish-purple varieties)
(B.) Reddish-purple
(C.) Black

FSC 8915

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

1.3 Grades. Canned berries shall be of the following grades as specified:

Grade A (Fancy)
Grade B (Choice)
Grade C (Standard)

1.4 Types of pack. When specified (see 6.1) "heavy pack" (high drained weight) blackberries and other similar berries are available for any grade, "in water", and in No. 10 cans.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specification:

PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of.

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specification, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

Military Specification:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of Military Specifications and Standards required by supplier in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

U.S. Department of Health, Education and Welfare

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

U.S. Department of Agriculture

United States Standards for Grades of Canned Blackberries and Other Similar Berries such as Boysenberries, Dewberries, and Loganberries.

United States Standards for Grades of Canned Blueberries.

United States Standards for Grades of Canned Raspberries.

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products.

U.S. Standard for Condition of Food Containers.

(Single copies may be obtained without charge from the Information Division, Consumer and Marketing Service, U.S. Department of Agriculture, Washington, D.C. 20250.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists.

(Copies may be obtained from the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

3. REQUIREMENTS

3.1 Material. Canned berries shall be prepared from stemmed, properly ripened, sound, sorted, cleaned and washed fresh berries, and unless otherwise specified, shall be of the latest season's crop.

3.2 Description of kinds, types, types of pack, and grades. Unless otherwise specified, the interpretation of requirements (descriptive terms) in 1.1, 1.2 and 1.3 shall be in accordance with the United States Standards for Grades of Canned Blackberries and Other Similar Berries, such as Boysenberries, Dewberries, and Loganberries; Canned Blueberries or Canned Raspberries, respectively (and lot inspection and certification procedures and tolerances applicable thereto as set forth in the Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products) in effect on the date of invitation for bids.

3.3 Sirup requirements. Unless otherwise specified, canned blackberries and other similar berries and raspberries, grade A quality, shall be packed in extra heavy sirup; grade [illegible]ty shall be packed in heavy sirup (see 1.2).

3.3.1 The Degree Brix of the liquid media. The degree Brix of the liquid media at the time of "cut-out" shall be as follows, when tested as specified in 4.3.2.

Blackberries, boysenberries, dewberries, loganberries, and youngberries.

Extra heavy sirup - 24° or more, but not more than 35° Brix.
Heavy sirup - 19° or more, but less than 24° Brix.
Light sirup - 14° or more, but less than 19° Brix.
Slightly sweetened water - Less than 14° Brix.
Water pack - Packed in water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

Blueberries.

Extra heavy sirup - 25° or more, but not more than 35° Brix
Heavy sirup - 20° or more, but less than 25° Brix.
Light sirup - 15° or more, but less than 20° Brix.
Slightly sweetened water - Less than 15° Brix.
Water pack - Packed in water.
(Blueberries are usually packed in water)

Raspberries.

Red or reddish purple types.

Extra heavy sirup or extra heavy raspberry juice - 28° Brix or more, but not more than 35° Brix.
Heavy sirup or heavy raspberry juice sirup - 22° or more, but less than 28° Brix.
Light sirup or light raspberry juice sirup - 14° or more, but less than 22° Brix.
Water pack - Packed in water.
In raspberry juice - Packed in raspberry juice.

Black types.

Extra heavy sirup or extra heavy raspberry juice sirup - 27° Brix or more, but not more than 35° Brix.
Heavy sirup or heavy raspberry juice sirup - 20° or more, but less than 27° Brix.
Light sirup or light raspberry juice sirup - 14° or more, but less than 20° Brix.
Water pack - Packed in water.
In raspberry juice - Packed in raspberry juice.

3.4 Drained weight. Drained weights shall be determined in accordance with the procedures and shall meet the recommended minimum drained weights specified in the applicable United States Standards for Grades.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. For Military agencies the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Samples of finished products covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished products may be taken at any time by the Government inspector to determine compliance with the requirements.

4.2 Inspection.

4.2.1 General. Inspection shall be made by the procuring agency or a duly authorized representative at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing, at any suitable point in transit, or after delivery to the point of destination.

4.2.2 Factory inspection. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.3 Final inspection. Unless otherwise specified, final inspection shall be made after delivery to the point of destination (see 6.3).

4.2.4 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the Military, see 4.2.7.2).

4.2.5 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to furnish an official inspection certificate issued by the U.S. Department of Agriculture certifying that the grade of the product supplied is as required by the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

4.2.5.1 Inspection records of the examination and test shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.6 Preacceptance inspection (Military Agencies). Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.2.7 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except as otherwise stated herein.

4.2.7.1 Component and material inspection. In accordance with 4.2.5, and 4.2.5.1 above, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.7.2 Examination of end item.

4.2.7.2.1 External examination of cans and jars for condition and labeling. Examinations shall be in accordance with the United States Standards for Condition of Food Containers.

4.2.7.2.2 Examination for kinds, types, types of pack, sirup Brix, grade and drained weight. Examinations shall be in accordance with the United States Standards for Grades of Canned Blackberries and other Similar Berries, Canned Blueberries, or Canned Raspberries, respectively, and Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and related Products, as applicable.

4.2.7.2.3 Examination for can vacuum and interior enamel of cans. Examinations shall be in accordance with the criteria in PPP-C-29.

4.2.7.2.4 Examination of shipping containers for cans. Examination of shipping containers and packing shall be in accordance with the criteria in PPP-C-29.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

4.2.7.2.5 Examination of unit loads. Examination of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

4.3 Tests.

4.3.1 Chemical analyses. Unless otherwise specified, chemical analyses, if required by the purchaser, shall be made in accordance with the methods of the Association of the Official Analytical Chemists in effect on the date of the invitations for bids or in accordance with methods that give equivalent results.

4.3.2. Determination of degree Brix of the sirup. The degree Brix of the sirup or liquid medium, as applicable, shall be determined at the time of "cut-out" by a Brix hydrometer or by any other method which gives equivalent results.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

5.1.1 Levels A and C, cans. The product shall be packaged in accordance with Levels A or C requirements of PPP-C-29, as specified (see 6.1).

5.1.2 Level C, glass. The product shall be packaged in glass containers in accordance with good commercial practice.

5.2 Packing. The product shall be packed in accordance with 5.2.1 or 5.2.2, as applicable.

5.2.1 Levels A, B, and C, cans. The product shall be packed in accordance with PPP-C-29 for the level specified. (See 6.1).

5.2.2 Level C, glass. The product, packaged in accordance with 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078, for the type and class of load specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

5.4 Labeling and marking.

5.4.1 Military agencies.

5.4.1.1 Cans. Cans shall be labeled in accordance with MIL-L-1497.

5.4.1.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.2 Civilian agencies.

5.4.2.1 Unit containers, cans, and glass. Labeling of the unit containers shall comply with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with the requirements of Federal Standard 123 and such other information as specified by individual agencies.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) The kind, type of pack and grade desired (see 1.2, 1.3 and 1.4).
(c) When "heavy-pack" in water in No. 10 cans of blackberries or other similar berries are desired (see 1.4).
(d) The type of sirup desired if other than provided for under 3.3. Berries for Military procurements are normally packed in water.
(e) Size and kind of packaging required (see 5.1).
(f) Applicable levels of packaging and packing (see 5.1.1 and 5.2.1).
(g) Type and class of unit, when unit loading is specified (see 5.3).

6.2 If samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

6.3 Destination inspection (civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.4 Supersession date. This specification supersedes Z-B-00421D (Army-GL) dated September 3, 1969, Z-B-421A dated March 22, 1954, Z-B-491D dated August 2, 1963, and Z-R-91B dated August 25, 1959.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

GSA
AGR
VA

☆ U. S. GOVERNMENT PRINTING OFFICE : 1971 O - 482-190 (3167)

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

U.S.MAIL

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Z-B-421E Berries, Canned

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF OCT 1 64 WHICH MAY BE USED.

Source: https://assist.dla.mil - Downloaded: 2026-04-20T14:08
Check the source to verify that this is the current version before use.